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Past Papers Nat 5 Chemistry

2024 Marking Scheme

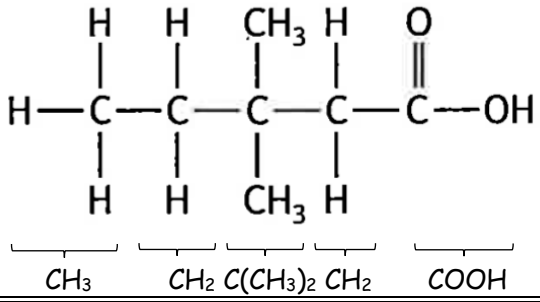
Grade Awarded	Mark Required		% candidates achieving grade
	(/125)	%	
A	+	%	%
B	+	%	%
C	+	%	%
D	+	%	%
No award	<	<%	%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	/25	/75	/25

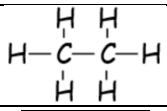
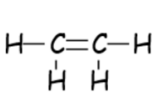
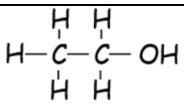
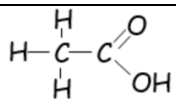
2024 National 5 Chemistry Marking Scheme

MC Qu	Answer	% Pupils Correct	Reasoning
1	B		$\text{Rate} = \frac{\Delta\text{Quantity}}{\Delta\text{Time}} = \frac{0.6\text{ bar}}{5\text{ min}} = 0.12\text{ bar min}^{-1}$
2	D		Isotopes have *same* number of protons but *different* number of neutrons ☒A W and X have same number of neutrons ☒B X and Z have different numbers of protons ☒C W and Y have different numbers of protons ☑D Y and Z have same number of protons but different numbers of neutrons
3	D		Atomic number of Sn = 50 ∴ no. of protons = 50 Mass number = no. of protons + no. of neutrons = 50 + 74 = 124
4	A		☑A tetraethylplumbane would be a covalent molecular substance as it is a liquid at room temperature and does not conduct in any state. ☒B a covalent network would be solid at room temperature with a high melting point ☒C an ionic lattice would be a solid at room temperature ☒D a metallic substance would conduct electricity in the liquid state
5	C		☒A The O atom is not directly bonded to a hydrogen atom ∴ no hydrogen bonding ☒B The N atom is not directly bonded to a hydrogen atom ∴ no hydrogen bonding ☑C The O atom is directly bonded to a hydrogen atom ∴ hydrogen bonding present ☒D The N atom is not directly bonded to a hydrogen atom ∴ no hydrogen bonding
6	B		☒A aluminium oxide is insoluble in water so there would be no change to the pH ☑B calcium oxide is soluble in water ∴ pH would increase as non-metal oxides dissolve to form alkaline solutions ☒C copper(II) oxide is insoluble in water so there would be no change to the pH ☒D zinc(II) oxide is insoluble in water so there would be no change to the pH
7	B		$2Mg + O_2 \longrightarrow 2MgO$ 2mol 1mol 2mol 4mol 2mol 4mol
8	D		☒A Nitrogen N_2 gfm = 2x14 = 28g ☒B Oxygen O_2 gfm = 2x16 = 32g ☒C Fluorine F_2 gfm = 2x19 = 38g ☑D Neon Ne gfm = 1x20 = 20g
9	C		☒A same number of moles of sodium hydroxide present after 25cm^3 water added ☒B dilution of alkali would lower pH as alkali is more dilute after 25cm^3 water added ☑C number of moles of sodium hydroxide present is constant and pH would be lower ☒D same number of moles of sodium hydroxide present after 25cm^3 water added
10	A		Alkanes are a homologous series of hydrocarbons which are flammable. Combustion of alkanes are exothermic reactions which would be suitable for use as a fuel.
11	C		Name: cyclooctyne / cyclononyne / cyclodecyne Number of carbons (n): n = 8 / n = 9 / n = 10 Hydrogens: 2n-4: (2x8)-4 = 16-4 = 12 / (2x9)-4 = 18-4 = 14 / (2x10)-4 = 20-4 = 16 Formula: C_8H_{12} / C_9H_{14} / $C_{10}H_{16}$
12	B		Pent-2-ene has formula C_5H_{10}. Isomers have *same formula* but *different structure*. ☒A Pentane has formula C_5H_{12} ∴ Different formula means not an isomer ☑B Pent-1-ene C_5H_{10} has same formula as pent-2-ene but different structure ☒C Molecule is pent-2-ene ∴ Same formula & same structure means not an isomer ☒D 2-methylbutane has formula C_5H_{12} ∴ Different formula means not an isomer

13 B



14 C

Name	Ethane	Ethene	Ethanol	Ethanoic acid
Structure				
Bonds Present	C–C ✓ C–O × C–H ✓ O–H ×	C–C × C–O × C–H ✓ O–H ×	C–C ✓ C–O ✓ C–H ✓ O–H ✓	C–C ✓ C–O ✓ C–H ✓ O–H ✓

15 B

☒A primary amine: 1 carbon atom bonded to the nitrogen atom
☑B secondary amine: 2 carbon atoms bonded to the nitrogen atom
☒C tertiary amine: 3 carbon atoms bonded to the nitrogen atom
☒D primary amine: 1 carbon atom bonded to the nitrogen atom

16 D

Fuel	Mass (kg)	Temperature (°C)	$E_h = cm\Delta T$
A	0.05	10	E_h = 4.18 x 0.05 x 10 = 2.09 kJ
B	0.05	20	E_h = 4.18 x 0.05 x 20 = 4.18 kJ
C	0.25	8	E_h = 4.18 x 0.25 x 8 = 8.36 kJ
D	0.25	18	E_h = 4.18 x 0.25 x 18 = **18.81 kJ**

17 A

☑A Increasing distance between burner and beaker would increase heat loss
☒B Placing a lid on the beaker would reduce heat loss to the surroundings
☒C Using a copper beaker would reduce heat loss travelling through glass beaker
☒D Using a heat/draft shield would reduce heat loss to the surroundings

18 A

Diagram shows metallic bonding with positive ions and delocalised electrons
☑A Metallic bonding: conducts electricity both as a solid and a liquid
☒B Covalent Molecular: low melting point and no conduction as solid or liquid
☒C Ionic Bonding: no conduction as solid but conduction as a liquid
☒D Covalent Network: high melting point and no conduction as soilid or liquid

19 B

Acid + Metal → Salt + Hydrogen
hydrochloric acid + magnesium → magnesium chloride + **hydrogen**

20 A

❶ Al^{3+} + $3e^-$ → Al
❷ $2O^{2-}$ → O_2 + $4e^-$

❶ x4 $4Al^{3+}$ + $12e^-$ → 4Al
❷ x3 $6O^{2-}$ → $3O_2$ + $12e^-$

add $4Al^{3+}$ + $6O^{2-}$ → 4Al + $3O_2$

21 B

☒A Ammonia has a formula of NH_3
☑B ammonia dissolves in water to form ammonium hydroxide which reacts with acids
☒C ammonia turns damp pH paper blue as ammonia dissolves in water for form alkali
☒D ammonia is a gas at room temperature (boiling point = -33°C)

22 D

☒A K_2CO_3 - 1 essential element for plant growth (potassium)
☒B K_2SO_4 - 1 essential element for plant growth (potassium)
☒C NH_4NO_3 - 1 essential element for plant growth (nitrogen)
☑D $(NH_4)_3PO_4$ - 2 essential elements for plant growth (nitrogen and phosphorus)

23	C		nitrogen + hydrogen $\xrightleftharpoons{\text{iron catalyst}}$ ammonia $N_2 + 3H_2 \xrightleftharpoons{\text{iron catalyst}} 2NH_3$
24	D		$^{228}_{88}Ra \xrightarrow{\beta} {}^{228}_{89}Ac \xrightarrow{\beta} {}^{228}_{90}Th \xrightarrow{\alpha} {}^{224}_{88}Ra$
25	D		☒A copper gives a flame test colour of blue-green ☒B copper gives a flame test colour of blue-green ☒C copper(II) Cu^{2+} ions are blue in solution ☑D copper(II) Cu^{2+} ions are blue in solution and a flame test colour of blue-green

2024 National 5 Chemistry Marking Scheme

Long Qu	Answer	Reasoning
1a(i)	One answer from:	$2H_2O_2 \longrightarrow 2H_2O + O_2$ or $H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$
1a(ii)	catalyst	A catalyst is a chemical which speeds up a chemical reaction without being used up ion th reaction. The same mass of catalyst is left at the end of the reaction.
1a(iii)A	Insoluble in water	For a gas to be collected over water by this method, the gas must be insoluble in water so the full volume of gas collects in the inverted measuring cylinder. Examples of soluble & insoluble gases include:

Soluble Gases	NO_2	NH_3	HCl	SO_2	(CO_2)
Insoluble Gases	N_2	O_2	noble gases	hydrocarbons	

CO_2 is slightly soluble in water so some CO_2 will dissolve in water (like fizzy drinks) but CO_2 can be collected by this method in some experiments. The best way to collect soluble gases is in a gas syringe.

1a(iii)B — 36

$$\text{Rate} = \frac{\Delta\text{Quantity}}{\Delta\text{Time}}$$

$$1.2\ cm^3\ s^{-1} = \frac{\Delta\text{Quantity}}{30\ s}$$

$$1.2\ cm^3\ s^{-1} \times 30s = \Delta\text{Quantity}$$

$$36cm^3 = \Delta\text{Quantity}$$

1a(iii)C — Less than 1.2

Reaction rates decrease as reactions proceed as the concentration of reactants decreases as the reactants get used up. There are less collisions between the lower concentration of reactants which lowers the reaction rate. The average reaction rate must be lower between 60 and 90 seconds than it was between 0 and 30 seconds.

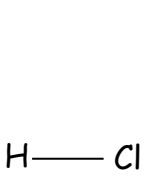
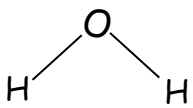
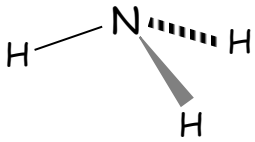
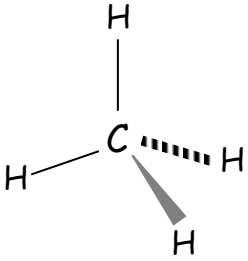
1b(i) — limewater turns cloudy/milky

Gas	Test for Gas
Carbon dioxide	turns limewater milky
Hydrogen	burns with a pop
Oxygen	relights a glowing splint

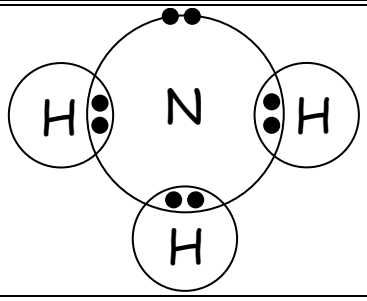
1b(ii) — endothermic

Type of Reaction	Definition
Exothermic	Reaction which releases (heat) energy
Endothermic	Reaction which takes in (heat) energy from the surroundings

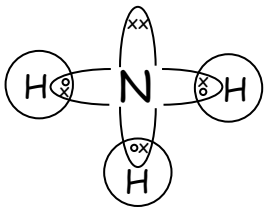
2a(i) — trigonal pyramidal

HCl	H_2O	NH_3	CH_4
			
Linear	Angular	Trigonal Pyramidal	Tetrahedral

2a(ii) — Answer showing:



or

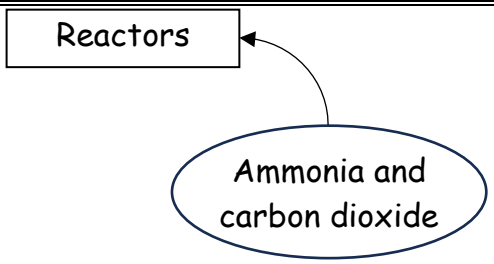


2b	Nitric Acid	The end product of the Ostwald Process is nitric acid. Ammonia NH_3 and Oxygen O_2 are the reactants which react to form nitrogen monoxide NO and water H_2O using a platinum catalyst. Nitrogen monoxide NO reacts with air and water to form nitric acid HNO_3 Heat is given off in this reaction and the heat supply to start the reaction can be turned off once the reaction has started.

2c — Graph showing:

1 mark	1 mark	1 mark	1 mark
One mark is awarded for a graph which shows points plotted rather than bars.	The axis/axes of the graph has/have suitable scale(s). • plotted points occupies at least half of the width and half of the height of the graph paper • The axes have suitable scales	The axes of the graph have suitable labels and units.	All data points plotted accurately (within a half box tolerance) with either a line of best fit drawn or plots joined by curve. This mark can only be accessed if linear scales for both axes have been provided.

2d — Arrow drawn as shown:



3a	Essential oils	Problem Solving: Information from a passage
3b	0.885 / 0.905 204 / 183	Problem Solving: Transfer of data from passage to table
3c	Lavender Spike Oil	Problem Solving: Information from a passage
3d	Unsaturated oils will decolourise bromine solution	Bromine solution is decolourised when unsaturated hydrocarbons with C=C double bonds are present. The bromine molecules in bromine solution add across the C=C double bonds in the essential oils and the bromine solution loses its yellow colour when all the bromine molecules add onto the unsaturated essential oil molecules
3e	Stronger intermolecular bonds	Geraniol has a higher boiling point than linalool and nerol because the geraniol. Molecules are closer to each other duen to stronger intermolecular forces/bonds/attractions between molecules.
3f	2.643	mass (g) = density (g cm^{-3}) x volume (cm^3) mass (g) = 0.881 g cm^{-3} x 3 cm^3 mass (g) = 2.643 g
4a	homologous series	A homologous series is a family of compounds with same/similar chemical properties and a general formula. Examples of homologous series include:

Homologous Series	Alkane	Alkene	Cycloalkane	Alcohol	Carboxylic acid
General Formula	C_nH_{2n+2}	C_nH_{2n}	C_nH_{2n}	$C_nH_{2n+1}OH$	$C_nH_{2n+1}COOH$

4b(i) — 141

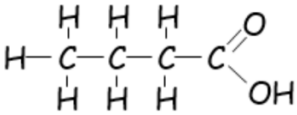
Name	methanol	ethanol	propan-1-ol	butan-1-ol	pentan-1-ol
Boiling Point (°C)	65	78	97	118	-
Difference (°C)		13	19	21	(23)
Prediction (°C)	-	-	-	-	141

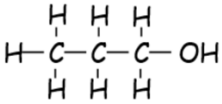
4b(ii)	Increasing in carbons increases the boiling point	The longer the carbon chain in the alcohols the higher the boiling point due to stronger intermolecular bonds/attractions/forces.

4c(i) — hydroxyl, carboxyl

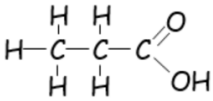
—O—H	—C(=O)—OH
hydroxyl group	carboxyl group

4c(ii)

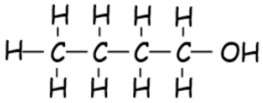




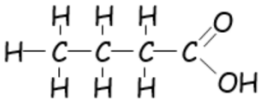
propan-1-ol



propanoic acid



butan-1-ol



butanoic acid

4c(iii)

$(K^+)_2Cr_2O_7^{2-}$

Write down valency below each element's symbol	Put in Cross-over Arrows	Follow arrows to get formula
K $Cr_2O_7^{2-}$	K Cr_2O_7	$K_2Cr_2O_7$
1 2	1 2	add charges
		$(K^+)_2Cr_2O_7^{2-}$

5

Open Question:

Candidate Marks			
3 marks	2 marks	1 mark	0 marks
%	%	%	%

3 mark answer	2 mark answer	1 mark answer
Demonstrates a **good understanding** of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a **reasonable understanding** of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a **limited understanding** of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.

6a

Delocalised electrons

Metals have metallic bonding where positive ions (nucleus and inner shells) are surrounded by delocalised electrons (in the outer shell). Those delocalised electrons can jump from outer shell to outer shell so electricity conduction can take place.

6b

Heat with carbon

Method	Electrolysis	Heat With Carbon	Heat Alone
Metals Made This Way	Potassium, Sodium, Lithium, Calcium, Magnesium, Aluminium	Zinc, Iron, Nickel, Tin, Lead, Copper	Mercury, Silver, Gold, Platinum
Reason	most reactive metals	medium reactive metals	least reactive metals

6c(i)

C
A
copper
D
B

Direction of electron Flow:	**Size of Voltage:**
Metals A and C must be above copper in electrochemical series as electrons flow from metals A and C to copper.	Metal C gives a bigger voltage with copper than metal A with copper does. Metal C is higher up electrochemical series than metal A
Metals B and D must be below copper in electrochemical series as electrons flow from copper to metals B and D.	Metal B gives a bigger voltage with copper than metal D with copper does. Metal B is lower down electrochemical series than metal D

6c(ii)

Glucose (is molecular and) contains no ions

Glucose is covalent because there are only non-metals in its formula $C_6H_{12}O_6$. To complete the circuit an ionic solution must be used as an electrolyte but covalent molecular glucose contains no ions to complete the circuit.

6d(i)

Ion/salt bridge

An ion bridge (or salt bridge) is filter paper soaked in an ionic solution.

6d(ii)

Any soluble lead compound:

One from: Lead bromide Lead chloride Lead nitrate

7a

Kills bacteria and viruses

Problem Solving: Information from a passage

7b

sodium, hydrogen, sulfur and oxygen

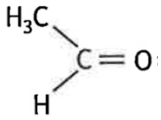
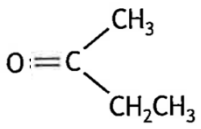
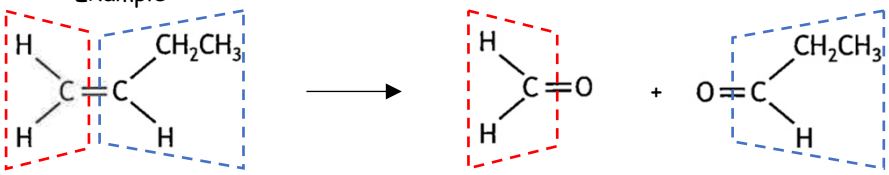
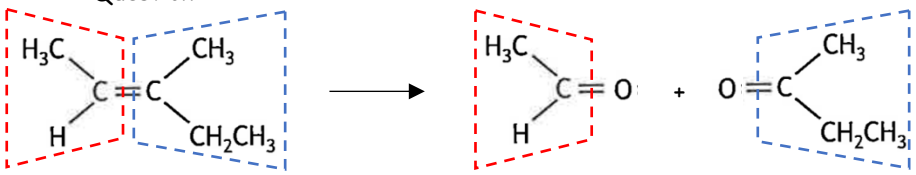
$\underbrace{\text{sodium}}_{\text{sodium}}\ \underbrace{\text{hydrogen}}_{\text{hydrogen}}\underbrace{\text{sul}}_{\text{sulfur}}\underbrace{\text{fite}}_{\text{oxygen}}$

7c

$Cl_2 + H_2O$
↓
$HOCl + HCl$

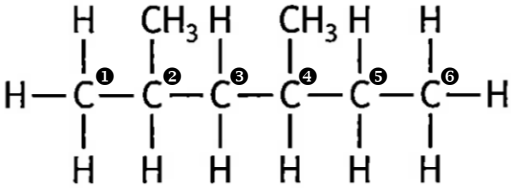
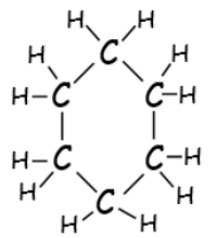
chlorine + water → hypochlorous acid + hydrochloric acid

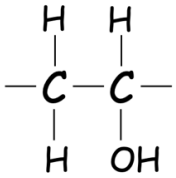
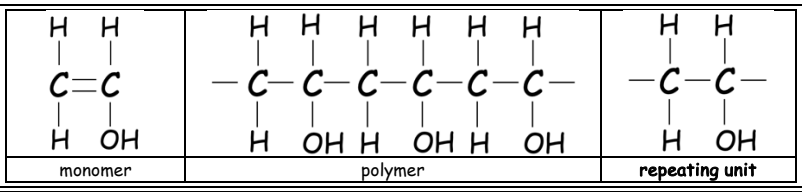
$Cl_2 + H_2O \rightarrow HOCl + HCl$

7d	Diagram showing:  and 	Example:  Question: 
8a(i)	Equation showing:	$Ag^+_{(aq)} + NO_3^-{}_{(aq)} + Na^+_{(aq)} + Cl^-_{(aq)} \rightarrow Ag^+Cl^-_{(s)} + Na^+_{(aq)} + NO_3^-{}_{(aq)}$ Identify Spectator Ions and cancel out Spectator Ions $Ag^+_{(aq)} + \cancel{NO_3^-{}_{(aq)}} + \cancel{Na^+_{(aq)}} + Cl^-_{(s)} \rightarrow Ag^+Cl^-_{(s)} + \cancel{Na^+_{(aq)}} + \cancel{NO_3^-{}_{(aq)}}$ Re-write equation without spectator ions $Ag^+_{(aq)} + Cl^-_{(aq)} \rightarrow Ag^+Cl^-_{(s)}$

8a(ii)	Precipitation	A precipitation reaction occurs when two ions in solution meet when two solutions are mixed. Some ion combinations form insoluble precipitates when their ions meet.
		• Take the names of the two original solutions
		  ○ *silver nitrate* + sodium chloride
		• Swap the names over between the two solutions
		  ○ *silver* chloride + sodium *nitrate*
		• Look up the data booklet for an insoluble precipitate that would form
		  ○ *silver* chloride is insoluble and sodium *nitrate* is soluble
		• Silver chloride is a precipitate which can be removed by filtration.

8b(i)	pipette	Pipettes are the most accurate way to transfer fixed volumes of solutions in a titration. They are more accurate than burettes, measuring cylinder and beakers.
8b(ii)	1st titre is inaccurate	The first titre, also known as the rough titre, is not designed to be accurate but to give an indication of where the endpoint of the reaction is. The experiment is repeated and the majority of the volume of reactant from the burette is added in one large addition and then the remaining volume is added drop by drop until the indicator changes colour. This repeated into two titres are produced within $0.2cm^3$.
8b(iii)	0.312	no. of mol = volume x concentration = 0.0156litre x 0.1mol l^{-1} = 0.00156mol $H_2C_2O_4 + 2NaOH \longrightarrow Na_2C_2O_4 + 2H_2O$ 1mol 2mol 0.00156mol 0.00312mol $\text{concentration} = \frac{\text{no. of mol}}{\text{volume}} = \frac{0.00312\text{mol}}{0.01\text{litres}} = 0.312 \text{ mol } l^{-1}$
8b(iv)	Pink to colourless	The conical flask contains sodium hydroxide at the beginning and the phenolphthalein indicator is pink in alkaline conditions. As oxalic acid is added from the burette into the conical flask, the sodium hydroxide is neutralised and the pH of the contents decreases until it reaches pH=7 when neutralisation occurs and the indicator changes colour to colourless.
9a(i)	No fizzing when more nickel carbonate is added	Nickel(II) carbonate reacts with sulfuric acid, carbon dioxide gas is produced in the reaction. The reaction mixture will stop fizzing when the nickel(II) carbonate runs out so more nickel(II) carbonate is added and it will start to fizz again. The process is repeated until adding more nickel(II) carbonate doesn't start fizzing in the reaction mixture indicating there is no acid left and neutralisation is complete.
9a(ii)	Water	All neutralisation reactions produce water. acid + metal oxide → salt + **water** acid + metal hydroxide → salt + **water** acid + metal carbonate → salt + **water** + carbon dioxide

9a(iii)	Base	A base is a chemical which neutralises an acid. There are three types of bases metal oxide | metal hydroxide | metal carbonate
9a(iv)	0.025	no. of mol = volume x concentration = 0.025litre x 1mol l^{-1} = 0.025mol
9b	filtration	Insoluble solids are removed from liquids by filtrations. • the insoluble solid is the residue in the filter paper • the liquid is the filtrate that goes through the filter paper and collects underneath.
9c	Heat solution with Bunsen burner	The water can be evaporated quickly by being heated by a Bunsen burner. The crystals are not as well formed but it does happen much quicker.
9d	37.9	gfm $NiSO_4$ = (1x58.5)+(1x32)+(4x16) = 58.5 + 32 + 64 = 154.5g $\%Ni = \frac{\text{mass of Nickel}}{\text{gfm}} \times 100 = \frac{58.5}{154.5} \times 100 = 37.9\%$
10a	Answer to include:	Compound containing the elements carbon and hydrogen only
10b(i)	3460	Bond | Bond energy (kJ) | Number of bonds made | Total Energy Released (kJ) C=O | 804 | 2 | 2x 804 = 1608 O–H | 463 | 4 | 4x 463 = 1852 Total energy to **make** all the bonds in the products = | = 3460
10b(ii)	- 816	Energy Change = Total energy to **break** all the bonds in thee reactants - Total energy to **make** all the bonds in the products Energy Change = 2644 - 3460 Energy Change = - 816 kJ
10c(i)	2,4-dimethylhexane	 2,4-dimethylhexane Sidegroups on C_2 & C_4 | 2x methyl -CH_3 sidegroups | 6 carbons on main chain | C-C single bonds in main chain
10c(ii)		Any C_6H_{12} structure contain a carbon ring is correct. Correct answers include: • cyclohexane • methylcyclopentane • 1,2-dimethylcyclobutane • 1,3-dimethylcyclobutane • 1,1,2-trimethylcyclopropane • 1,2,3-trimethylcyclopropanee
10d(i)A	C_6H_{14}	octane → ethene + alkane X C_8H_{18} → C_2H_4 C_6H_{14} ∴ hexane
10d(i)B	hydration	Addition reactions add a molecule across a C=C double bond. • addition of water across C=C double bond is a **hydration** reaction • addition of hydrogen across C=C double bond is a **hydrogenation** reaction The following molecules are all able to add across a C=C double bond: H_2 | F_2 | Cl_2 | Br_2 | I_2 | H_2O | HF | HCl | HBr | HI Addition reaction gives one product | Addition reaction usually gives two product
10d(ii)A	poly(ethenol)	Monomer | ethene | propene | chloroethene | phenylethene | ethenol Polymer | poly(ethene) | poly(propene) | poly(chloroethene) | poly(phenylethene) | poly(ethenol)

10d(ii)B		 monomer / polymer / **repeating unit**
11	Open Question: Candidate Marks 3 marks: % 2 marks: % 1 mark: % 0 marks: %	**3 mark answer**: Demonstrates a **good understanding** of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem. **2 mark answer**: Demonstrates a **reasonable understanding** of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood. **1 mark answer**: Demonstrates a **limited understanding** of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.